9. A light source of wavelength 520 nm emits 1.04×10^{15} photons per second while the second source of 460 nm produces 1.38×10^{15} photons per second. Then the ratio of power of second source to that of first source is

a) 1.00 b) 1.02
c) 1.5 d) 0.98

10. If the mean wavelength of light from sun is taken as 550 nm and its mean power as 3.8×10^{26} W, then the average number of photons received by the human eye per second from sunlight is of the order of

a) 10^{45} b) 10^{42}
c) 10^{54} d) 10^{51}

11. The threshold wavelength for a metal surface whose photoelectric work function is 3.313 eV is

a) 4125 Å b) 3750 Å
c) 6000 Å d) 2062.5 Å

12. A light of wavelength 500 nm is incident on a sensitive metal plate of photoelectric work function 1.235 eV. The kinetic energy of the photo electrons emitted is (Take $h = 6.6 \times 10^{-34}$ Js)

a) 0.58 eV b) 2.48 eV
c) 1.24 eV d) 1.16 eV

13. Photons of wavelength λ are incident on a metal. The most energetic electrons ejected from the metal are bent into a circular arc of radius R by a perpendicular magnetic field having magnitude B. The work function of the metal is (KVPY-SX 2016)

a) $\frac{hc}{\lambda} - m_e + \frac{e^2B^2R^2}{2m_e}$

b) $\frac{hc}{\lambda} + 2m_e\left[\frac{eBR}{2m_e}\right]^2$

c) $\frac{hc}{\lambda} - m_ec^2 - \frac{e^2B^2R^2}{2m_e}$

d) $\frac{hc}{\lambda} - 2m_e\left[\frac{eBR}{2m_e}\right]^2$

14. The work functions for metals A, B and C are 1.92 eV, 2.0 eV and 5.0 eV respectively. The metal/metals which will emit photoelectrons for a radiation of wavelength 4100 Å is/are

a) A only
b) both A and B
c) all these metals
d) none

15. Emission of electrons by the absorption of heat energy is called……emission.

a) photoelectric
b) field
c) thermionic
d) secondary

Answers

1. d	**2.** c	**3.** d	**4.** d	**5.** a
6. d	**7.** b	**8.** b	**9.** c	**10.** a
11. b	**12.** c	**13.** d	**14.** b	**15.** c

II Short Answer Questions

1. Why do metals have a large number of free electrons?

2. Define work function of a metal. Give its unit.

3. What is photoelectric effect?

4. How does photocurrent vary with the intensity of the incident light?

5. Give the definition of intensity of light according to quantum concept and its unit.

6. How will you define threshold frequency?

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